Yaohui Shu

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Ph.D. candidate in Applied Mathematics specializing in Physics-Informed ML (SciML), Causal Inference, and LLM-based AI agents. Experienced in developing high-dimensional stochastic models and autonomous systems to solve complex physical and computational challenges in Python and C++.

Skills

ML & Stats: Causal inference, randomized experiments, time series analysis, hypothesis testing, PyTorch, JAX, LLM-based agents
Stochastic & Math: SDEs, Itô calculus, Monte Carlo simulations, Navier-Stokes/PDE solvers, pseudo-spectral methods, MPI, HPC
Programming: Python (NumPy, SciPy, pandas, scikit-learn), C++, R, MATLAB, Git, Linux, Bash

Education

Stony Brook University — Expected Dec 2027
Ph.D., Applied Mathematics and Statistics (GPA: 3.78/4.0) — Stony Brook, NY

- **Dissertation:** Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods

University of Rochester — Aug 2021 – Aug 2022
M.S., Finance (GPA: 3.81/4.0) | Certificate in Quantitative Finance (CQF) — Rochester, NY

Zhongnan University of Economics and Law / University of Rhode Island — 2017 – 2021
Dual Degree: B.A. in Finance & B.S.B.A. (GPA: 3.98/4.0) — China / RI

Research & Professional Experience

Stony Brook University — Aug 2023 – Present
Doctoral Researcher, Causal Inference and Computational Applied Mathematics — Stony Brook, NY

- **Causal Learning on Turbulence:** Formulated a randomized perturbation framework casting fluid dynamics as RCTs. Labeled all 16 causal links in 2D turbulence; confidence intervals covered the ground truth in 127/128 validation runs.
- **Pseudo-Spectral Solvers:** Developed high-performance 2D/3D Navier-Stokes and thermal convection solvers in Python/C++, validated against exact analytical solutions and linear stability theories.
- **LLM-based Simulation Agents:** Architected an autonomous AI agent for FronTier fluid simulations. Used expected information gain to dynamically propose hypotheses and schedule runs, reaching a 0.02% verification error on control benchmarks.
- **Advanced Causal Discovery:** Derived closed-form causal strength under feedback loops and hidden drivers via Girsanov's theorem. Developed and open-sourced an entropy-based causal discovery Python library.

Brookhaven National Laboratory (U.S. Department of Energy) — Jan 2024 – Jan 2026
Graduate Research Assistant, Stochastic Modeling and Information Flow — Upton, NY

- **DOE Simulation Lead:** Designed stochastic models, custom Monte Carlo routines, and PDE solvers for microphysical aerosol-cloud interactions, serving as the core developer of a high-priority DOE project.
- **First-Author Publication (GRL):** Introduced memory to Brownian-motion droplet growth models and measured causal information flow; verified analytical derivations against simulations of 100,000 droplets to recover causal orientations.
- **Theoretical Physics Formulation:** Reconciled maximum-entropy and Langevin equations of droplet-size distributions, deriving exact Fokker-Planck solutions to demonstrate the dominance of boundary conditions.

Stony Brook University — Aug 2026 – Present
Graduate Teaching Assistant, AMS 595: Fundamentals of Computing — Stony Brook, NY

- Instructed lab sessions and graded projects for graduate-level Python and C++ scientific computing courses.

Publications & Presentations

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., & Li, X. (2026). Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. [Code]
- **In Preparation:** Submissions for *Physical Review E*, *Journal of Atmospheric Sciences*, and two papers for *ICML 2027*.
- **Presentations:** First-author poster at *AGU Fall Meeting 2025*; co-author at *106th AMS Annual Meeting 2026*.